

Pythagorean Theorem and Coordinate Distance

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 13	4. 8	7. 0, 12	10. 24
2. 15	5. 10.8	8. 25	11. 10, -14
3. 13	6. 15	9. 7	12. 170

1. Ramp: rise 5, run 12, find the slanted surface. The ramp surface is the hypotenuse — it is opposite the right angle formed by the ground and the vertical rise.

$$c^2 = 12^2 + 5^2 = 144 + 25 = 169, \quad c = \sqrt{169}$$

13 ft

2. Ladder: hypotenuse 17, one leg 8, find the other leg. The ladder itself is the longest side, so 17 goes in the c slot and the height h is a leg.

$$8^2 + h^2 = 17^2 \Rightarrow h^2 = 289 - 64 = 225, \quad h = \sqrt{225}$$

15 ft

Grading note: a student who writes $\sqrt{17^2 + 8^2} \approx 18.79$ has put the ladder in a leg slot. Ask which side is longest — the answer can never exceed the ladder's own length.

3. Library (2, 3) to pool (7, 15). The horizontal gap is $7 - 2 = 5$ and the vertical gap is $15 - 3 = 12$; these are the legs of a right triangle whose hypotenuse is the straight-line distance.

$$d^2 = 5^2 + 12^2 = 25 + 144 = 169, \quad d = \sqrt{169}$$

13 units

Grading note: $5 + 12 = 17$ is the walking-along-the-streets distance, not the straight-line distance. Sketching the triangle prevents it.

4. Garden 24 m by 10 m: two sides vs. the diagonal. Two steps. First the diagonal: $d = \sqrt{24^2 + 10^2} = \sqrt{576 + 100} = \sqrt{676} = 26$ m. Then compare with the two-side walk: $24 + 10 = 34$ m. The saving is $34 - 26$.

8 m

5. Boats: 6 km north and 9 km east. North and east are perpendicular, so the two paths are the legs and the gap between the boats is the hypotenuse.

$$d = \sqrt{6^2 + 9^2} = \sqrt{36 + 81} = \sqrt{117} = 10.816\dots$$

Rounded to the nearest tenth: 10.8 km

Grading note: $6 + 9 = 15$ km is the distance if the boats sailed one after the other along the same bent path, not the direct gap.

6. Drone $(-4, -2)$ to $(5, 10)$. Horizontal change: $5 - (-4) = 9$. Vertical change: $10 - (-2) = 12$. Subtracting a negative is where most errors happen here — count the units on a sketch to confirm.

$$d = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$$

15 units

7. Points on the x -axis exactly 10 from $(6, 8)$. A point on the trail is $(x, 0)$. Its distance to $(6, 8)$ uses the same two legs as always: $|x - 6|$ horizontally and 8 vertically.

$$\begin{aligned}(x - 6)^2 + (0 - 8)^2 &= 10^2 \\(x - 6)^2 + 64 &= 100 \\(x - 6)^2 &= 36 \\x - 6 &= 6 \quad \text{or} \quad x - 6 = -6\end{aligned}$$

Both answers are real places on the trail: one on each side of the point directly below the station. $x = 0$ or $x = 12$

8. Kite: string 65 ft, height 60 ft. The taut string is the hypotenuse; the height is one leg and the ground distance g is the other.

$$g^2 = 65^2 - 60^2 = 4225 - 3600 = 625, \quad g = \sqrt{625}$$

25 ft

Grading note: $\sqrt{65^2 + 60^2} \approx 88.46$ means the string was treated as a leg. The ground distance must be shorter than the string.

9. Ladder slipping from 12 ft up to 5 ft up. The ladder's length never changes, so run the theorem twice with $c = 13$.

$$\begin{aligned}\text{before: } b_1 &= \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5 \\ \text{after: } b_2 &= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12\end{aligned}$$

The base moved $12 - 5$. 7 ft

Note the symmetry: the 5–12–13 triangle simply swapped its legs, which is why the numbers repeat.

10. Closed route $(1, 2) \rightarrow (7, 10) \rightarrow (7, 2) \rightarrow (1, 2)$. Three legs, and only the first one is slanted.

$$(1, 2) \rightarrow (7, 10) : \sqrt{(7-1)^2 + (10-2)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

$$(7, 10) \rightarrow (7, 2) : \text{vertical, } 10 - 2 = 8$$

$$(7, 2) \rightarrow (1, 2) : \text{horizontal, } 7 - 1 = 6$$

Total: $10 + 8 + 6$. 24 units

Grading note: vertical and horizontal legs need no square roots — just the difference of the changing coordinate. Using the distance formula there is correct but slower.

11. Points on $x = 8$ exactly 13 from $(3, -2)$. A point on the line is $(8, y)$, so the horizontal leg is fixed at $8 - 3 = 5$.

$$5^2 + (y + 2)^2 = 13^2$$

$$(y + 2)^2 = 169 - 25 = 144$$

$$y + 2 = 12 \quad \text{or} \quad y + 2 = -12$$

$y = 10$ or $y = -14$

The two points are $(8, 10)$ and $(8, -14)$ — one above the tower's height and one below, symmetric about $y = -2$.

12. Crate $120 \times 90 \times 80$ cm: the longest pole. Apply the theorem twice, exactly as the hint says.

$$\text{floor diagonal: } f = \sqrt{120^2 + 90^2} = \sqrt{14400 + 8100} = \sqrt{22500} = 150$$

$$\text{corner to corner: } d = \sqrt{f^2 + 80^2} = \sqrt{22500 + 6400} = \sqrt{28900}$$

The floor diagonal and the vertical edge meet at a right angle, which is what licenses the second use of the theorem. 170 cm

Grading note: stopping at 150 cm answers a different question — the longest pole that lies flat on the floor of the crate.